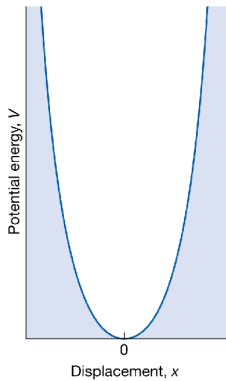


1-D HO : $V(x) = \frac{1}{2}\mu\omega^2x^2$



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use power series $v = \sum_{j=0}^{\infty} a_j \xi^j$

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use power series $v = \sum_{j=0}^{\infty} a_j \xi^j \implies \sum_j \xi^j \{(j+1)(j+2)a_{j+2} - 2ja_j + \lambda a_j\} = 0$

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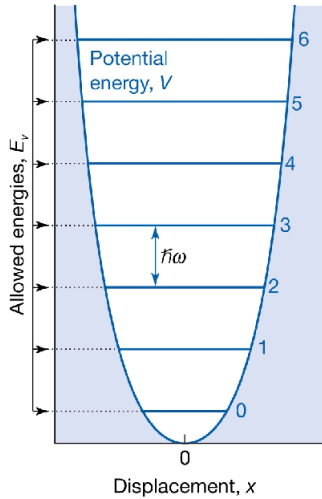
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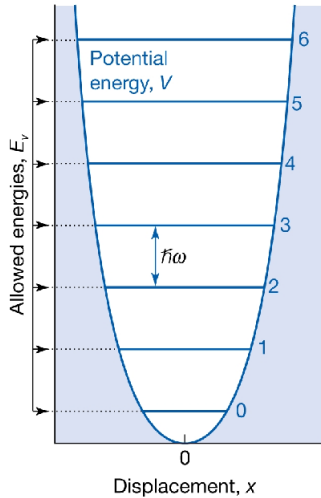
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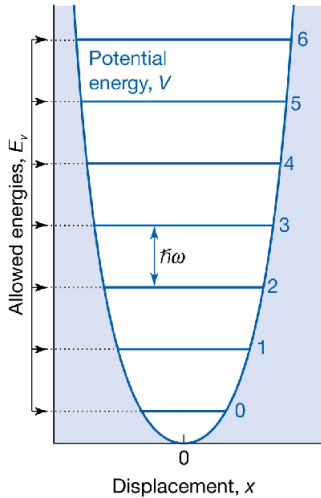
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$$\lambda = \frac{2E}{\hbar\omega} - 1, \quad \therefore E = \left(n + \frac{1}{2}\right) \hbar\omega$$



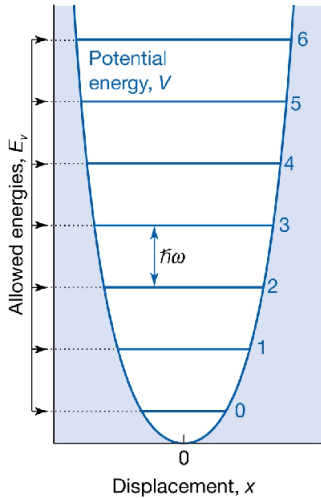


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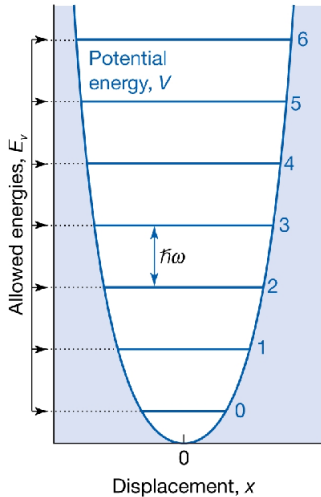
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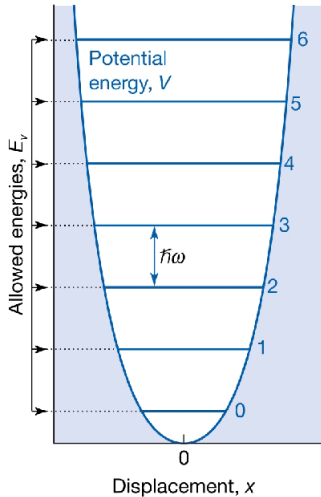


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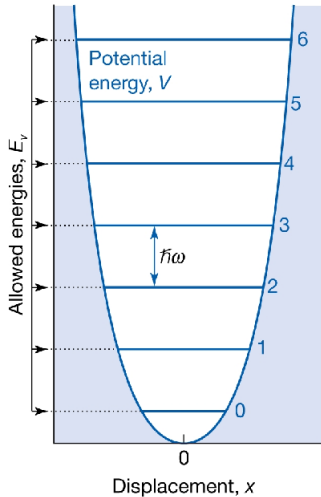
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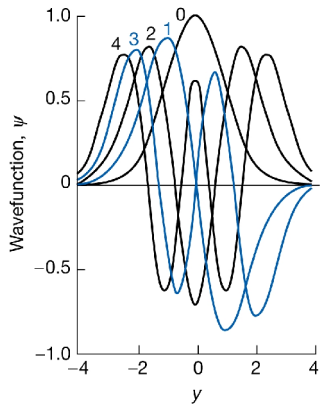
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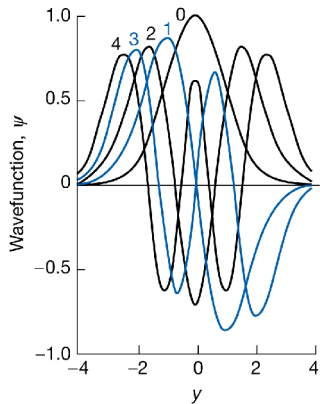
$$\nu = \frac{\Delta E}{\hbar} = 9 \times 10^{13} \text{ Hz} \equiv 3000 \text{ nm}$$

eigenfunctions

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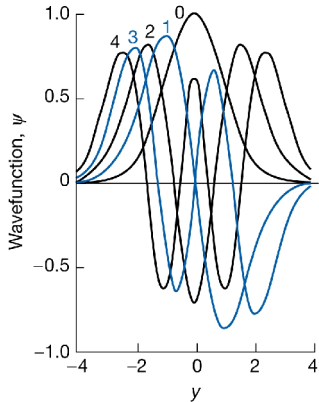


eigenfunctions



$$u(x) = N_n H_n(x) e^{-\frac{x^2}{2x_0^2}}$$

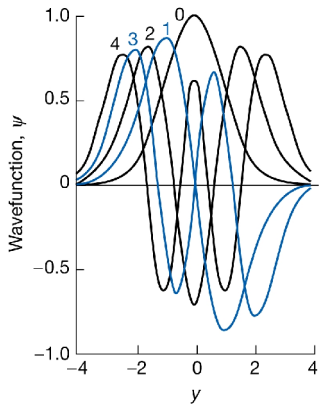
eigenfunctions



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eigenfunctions



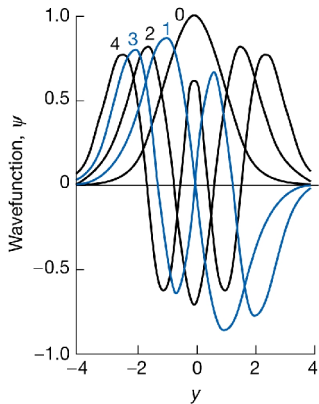
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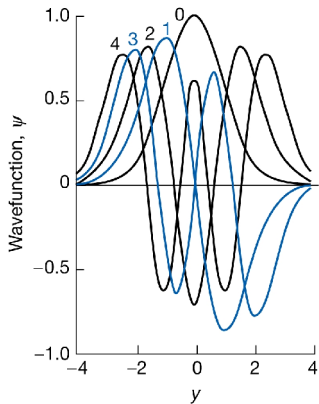
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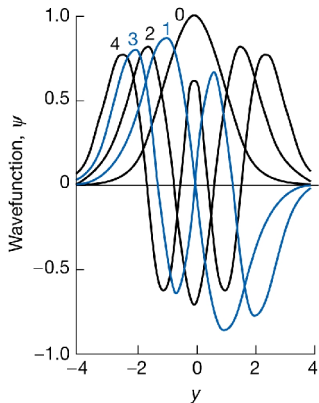
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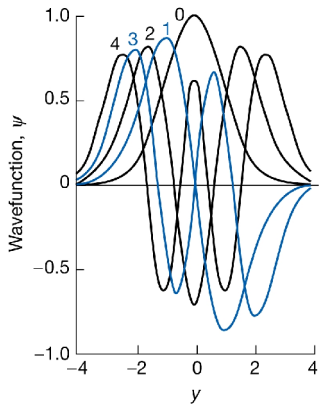
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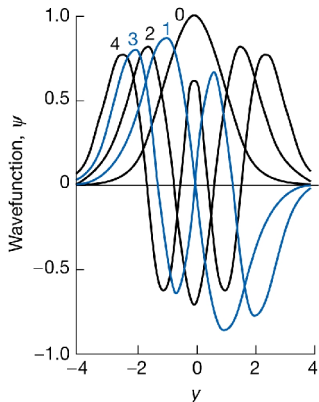
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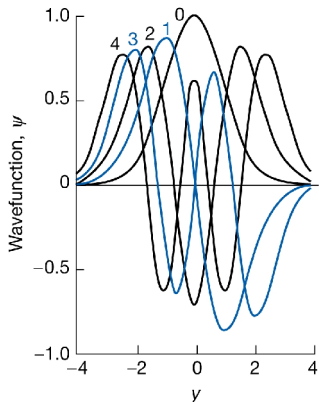
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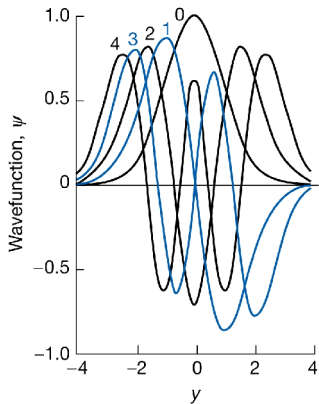
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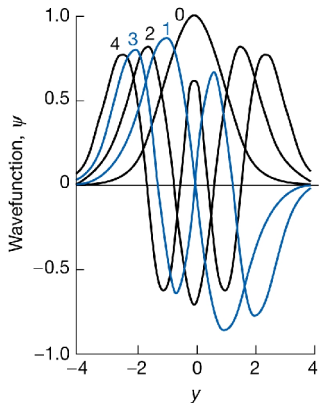
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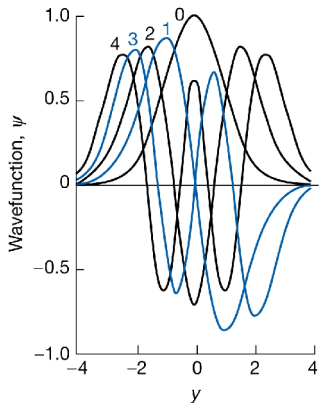
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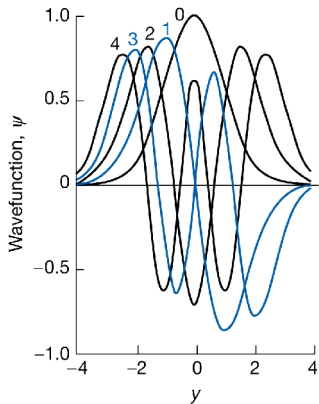
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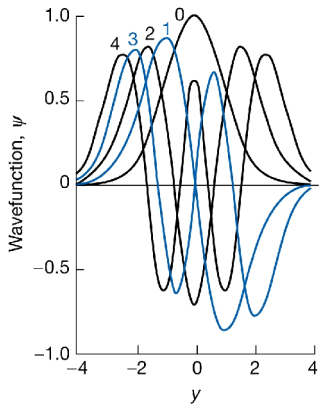
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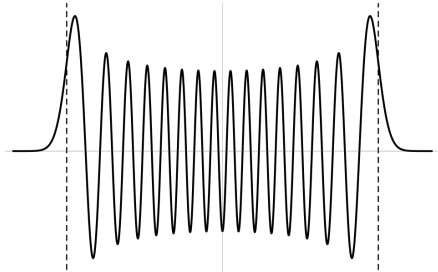
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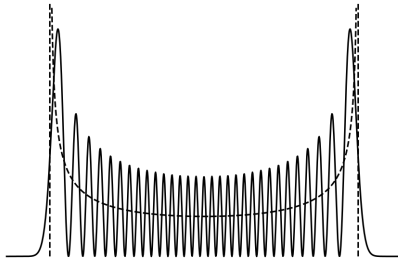
$$\text{and } u(\xi) = (-2\xi^2 + 1) e^{-\frac{\xi^2}{2}}, \quad H_2(\xi) = 4\xi^2 - 2$$

en.wikipedia.org/wiki/Quantum_harmonic_oscillator#/media/File:QuantumHarmonicOscillatorAn

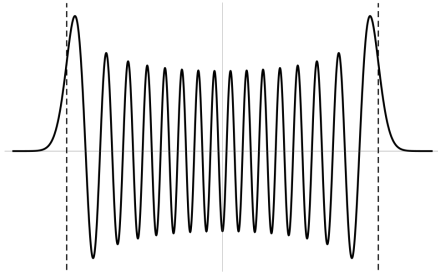
$n = 30$ wavefunction



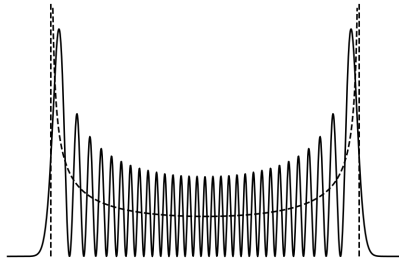
prob. density



$n = 30$ wavefunction



prob. density



system spends considerable time at the classical turning point

Method of second quantization

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 &\left(\sqrt{\frac{\mu\omega^2}{2}} \hat{x} - \frac{i\hat{p}}{\sqrt{2\mu}} \right) \left(\sqrt{\frac{\mu\omega^2}{2}} \hat{x} + \frac{i\hat{p}}{\sqrt{2\mu}} \right) - i\sqrt{\frac{\mu\omega^2}{2}} \sqrt{\frac{1}{2\mu}} (\hat{x}\hat{p} - \hat{p}\hat{x}) \\
 \hat{H} &= \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2 = \underbrace{\left(\sqrt{\frac{\mu\omega^2}{2}} \hat{x} - \frac{i\hat{p}}{\sqrt{2\mu}} \right)}_{\sqrt{\hbar\omega}\hat{a}^\dagger} \underbrace{\left(\sqrt{\frac{\mu\omega^2}{2}} \hat{x} + \frac{i\hat{p}}{\sqrt{2\mu}} \right)}_{\sqrt{\hbar\omega}\hat{a}} - i\frac{\omega}{2} \underbrace{[\hat{x}, \hat{p}]}_{i\hbar}
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label u by σ and write it as u_σ

More relations :

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one less than for u_σ

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for any polynomial, $P(\xi)$,

$$\frac{d}{d\xi} \left(P(\xi) e^{-\frac{\xi^2}{2}} \right) = e^{-\frac{\xi^2}{2}} \frac{d}{d\xi} \left(P(\xi) e^{-\frac{\xi^2}{2}} \right) - \xi e^{-\frac{\xi^2}{2}} P(\xi) e^{-\frac{\xi^2}{2}}$$

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$$\left(\frac{d}{d\xi} - \xi \right) P(\xi) e^{-\frac{\xi^2}{2}} = e^{-\frac{\xi^2}{2}} \cdot \underbrace{e^{\xi^2} \frac{d}{d\xi} \left(P(\xi) e^{-\xi^2} \right)} = Q(\xi) e^{-\frac{\xi^2}{2}}$$

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$$\text{For } n = 1, P(\xi) = 1 \text{ and } \left(\frac{d}{d\xi} - \xi \right) P(\xi) e^{-\frac{\xi^2}{2}} = e^{\xi^2} \left(\frac{d}{d\xi} e^{-\xi^2} \right) e^{-\frac{\xi^2}{2}}$$

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$$u_n = \frac{1}{(\pi x_0^2)^{\frac{1}{4}}} \frac{(-1)^n}{\sqrt{n!}} \frac{1}{\sqrt{2^n}} \left(\frac{d}{d\xi} - \xi \right)^n e^{-\frac{\xi^2}{2}} = N_0 (-1)^n \left(\frac{d}{d\xi} - \xi \right)^n e^{-\frac{\xi^2}{2}}$$

for any polynomial, $P(\xi)$,

$$\frac{d}{d\xi} \left(P(\xi) e^{-\frac{\xi^2}{2}} e^{-\frac{\xi^2}{2}} \right) = e^{-\frac{\xi^2}{2}} \frac{d}{d\xi} \left(P(\xi) e^{-\frac{\xi^2}{2}} \right) - \xi e^{-\frac{\xi^2}{2}} P(\xi) e^{-\frac{\xi^2}{2}}$$

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$$u_2 = N_2 (-1)^2 e^{\xi^2} \frac{d^2}{d\xi^2} \left(e^{-\xi^2} \right) e^{-\frac{\xi^2}{2}} = N_2 (4\xi^2 - 2) e^{-\frac{\xi^2}{2}}$$

Continuing $u_n = N_n H_n(\xi) e^{-\frac{\xi^2}{2}}$

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Recursion relations for Hermite polynomials:

$$H_n'' - 2\xi H_n' + 2n H_n = 0$$

$$H_{n+1} = 2\xi H_n - 2n H_{n-1}$$

orthogonality relation: $\int_{-\infty}^{\infty} e^{-\xi^2} H_n H_{n'} d\xi = \begin{cases} 0 & \forall n' \neq n \\ \sqrt{\pi} 2^n n! & \forall n' = n \end{cases}$

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expectation values:

$$\langle x \rangle = 0; \quad \langle x^2 \rangle = \left(n + \frac{1}{2}\right) \hbar \sqrt{\mu k}; \quad \langle V \rangle = \frac{1}{2} k x^2 = \frac{1}{2} \left(n + \frac{1}{2}\right) \hbar \omega$$

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Probability of locating the system in ground state outside the well:

orthogonality relation:
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Probability of locating the system in ground state outside the well:

$$P = 2 \int_{x_{t.pt.}}^{\infty} \psi_0^2 dx = 2x_0 N_0^2 \int_1^{\infty} \left(e^{-\frac{\xi^2}{2}}\right)^2 d\xi$$

orthogonality relation:
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Probability of locating the system in ground state outside the well:

$$\begin{aligned} P &= 2 \int_{x_{t.pt.}}^{\infty} \psi_0^2 dx = 2x_0 N_0^2 \int_1^{\infty} \left(e^{-\frac{\xi^2}{2}}\right)^2 d\xi \\ &= \sqrt{\frac{2}{\pi}} \int_1^{\infty} e^{-\xi^2} d\xi = 1 - \operatorname{erf}(1) = 1 - 0.843 = 0.157 \end{aligned}$$